



Inspiring Excellence

Geometry-Aware Multimodal Fusion for Surface Defect Detection

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ABSTRACT

Multimodal industrial defect detection is critical for manufacturing precision but is often **compromised when sensors degrade** in volatile environments. To solve this, a resilient **deep learning framework** was developed to **ensure inspection integrity** using **RGB, 3D point clouds, and semantic anchors**. The proposed architecture brings a key innovation by integrating a "Sensor Sentinel" for proactive reliability forecasting and "Geometric Feature Injection," outperforming standard reactive models. To **overcome the "geometry-blindness"** of Vision Transformers and build physical robustness, this research integrated explicit **surface curvature encoding**, proving the model maintains accuracy during sensor failures, establishing it as a self-aware tool for Industry 4.0.

PROBLEM STATEMENT

Traditional industrial inspection fails when sensors degrade, while **current AI models are often reactive "blind" systems** that break under uncertainty. To solve this, we developed a resilient framework built on four key contributions:

- 1. Reactive Nature:** Current systems only address missing data after complete sensor failure, ignoring progressive degradation ("soft failures") like noise or blur, which leads to significant accuracy drops during operation.
- 2. Geometry-Blindness:** Standard Vision Transformers treat 3D data as raw coordinates lacking explicit understanding of surface curvature, causing models to fail when visual texture is unreliable or low-contrast.
- 3. Geometry-Aware Design:** Sensor health monitoring is isolated from detection logic, allowing models to confidently trust data from drifting or uncalibrated sensors.

LITERATURE REVIEW

- Robustness in Missing Modalities: Cross-modal alignment is essential for maintaining performance during sensor failures [1].
- Prompt-Based Learning: Learnable prompts allow frozen foundation models to adapt to specialized downstream tasks efficiently [2].
- Geometric Context: Incorporating surface curvature and normals is crucial for accurate 3D defect detection [3].

KEY FINDING: WHY MISDD-MM MODEL BETTER

ROBUSTNESS (Sensor Sentinel)
88.7% Accuracy

Maintains high detection performance even when 80% of sensor data is missing or noisy ($\eta=0.8$).

GEOMETRY (Shape Logic)
Geometry-Aware

Uses Gaussian Curvature & Surface Normals to detect defects when visual texture is unreliable.

EFFICIENCY (Ultra-Lightweight)
13.1 ms Latency

Adds only 0.17% extra parameters. Runs in real-time on standard hardware (RTX 4090).

DATASET & PREPROCESSING

Dataset: Industrial dataset comprising 15,629 multimodal samples (RGB images and 3D point clouds) across 15 categories, augmented with sensor reliability labels and geometric surface data.

Feature Preprocessing:

- 3D Outlier Removal:** Applied to point clouds via Connected Components to filter isolated sensor noise.
- Geometric Extraction:** Surface normals and curvatures are computed to provide a structural inductive bias.

Image Preprocessing: The image illustrates the preprocessing of 3D sensor data by separating x, y, and z coordinate maps for geometric anomaly analysis.

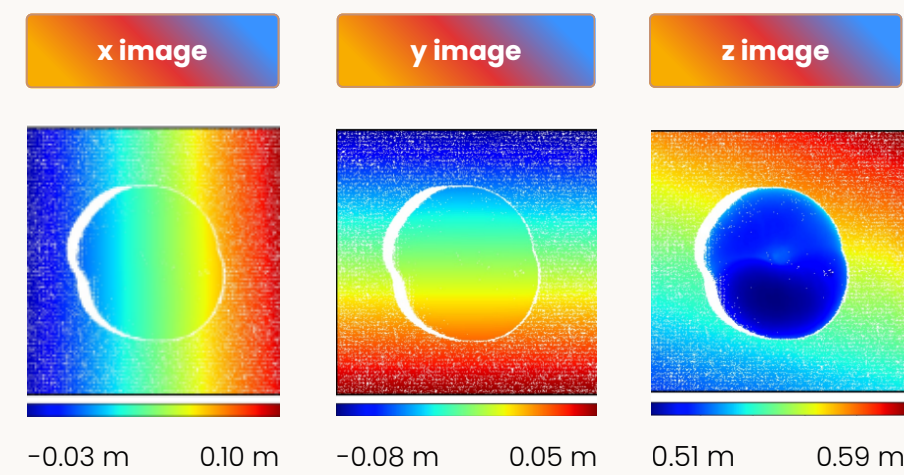


Fig-1: Preprocessing.

MODEL ARCHITECTURE

The architecture features a novel, **prompt-based framework that uses geometric feature injection to extract structural patterns** from 3D point clouds, intelligently fusing them with RGB data via a proactive sensor sentinel for a single, resilient prediction.

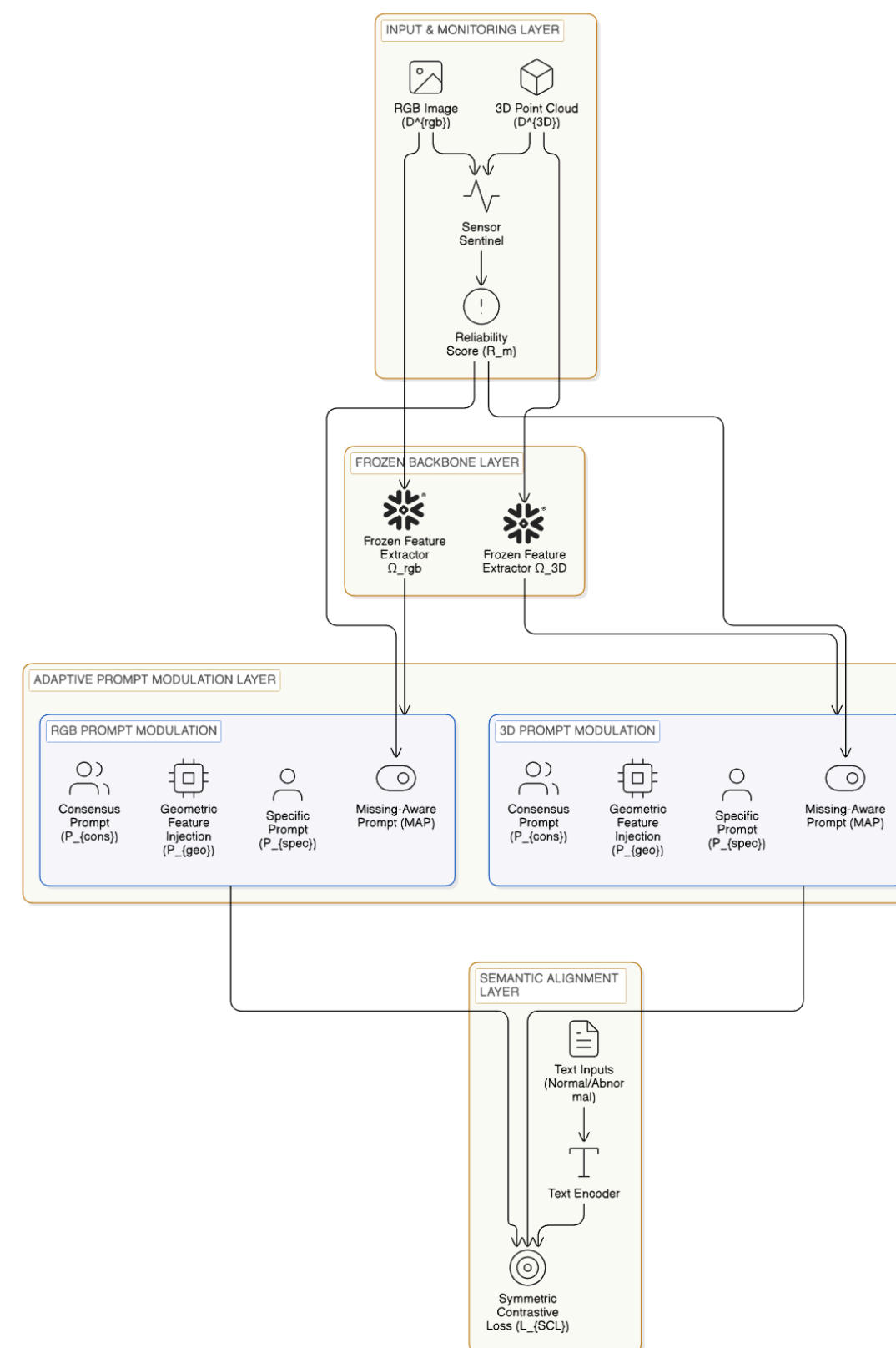


Fig-2: MISDD-MM Architecture.

TRAINING

The training process employs a **missing-rate simulation ($\eta=0.3$)**, with **100 epochs** and a **batch size of 32**. Finally, the symmetric contrastive loss is minimized using the AdamW optimizer to select the optimal model.

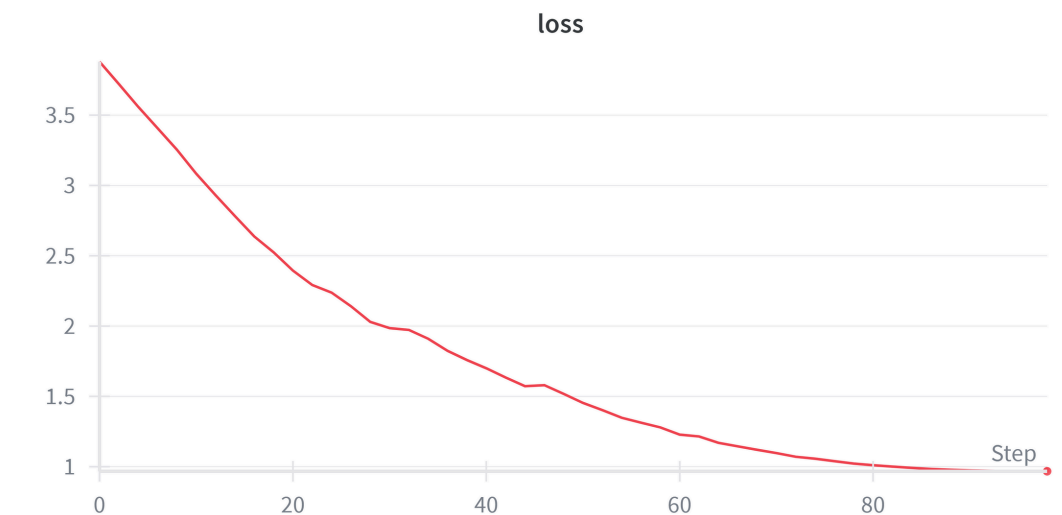


Fig-3: Loss vs Epoch

RESULTS & ANALYSIS

Bar Chart: MISDD-MM Performance Resilience Across Variable Missing Rates (η).

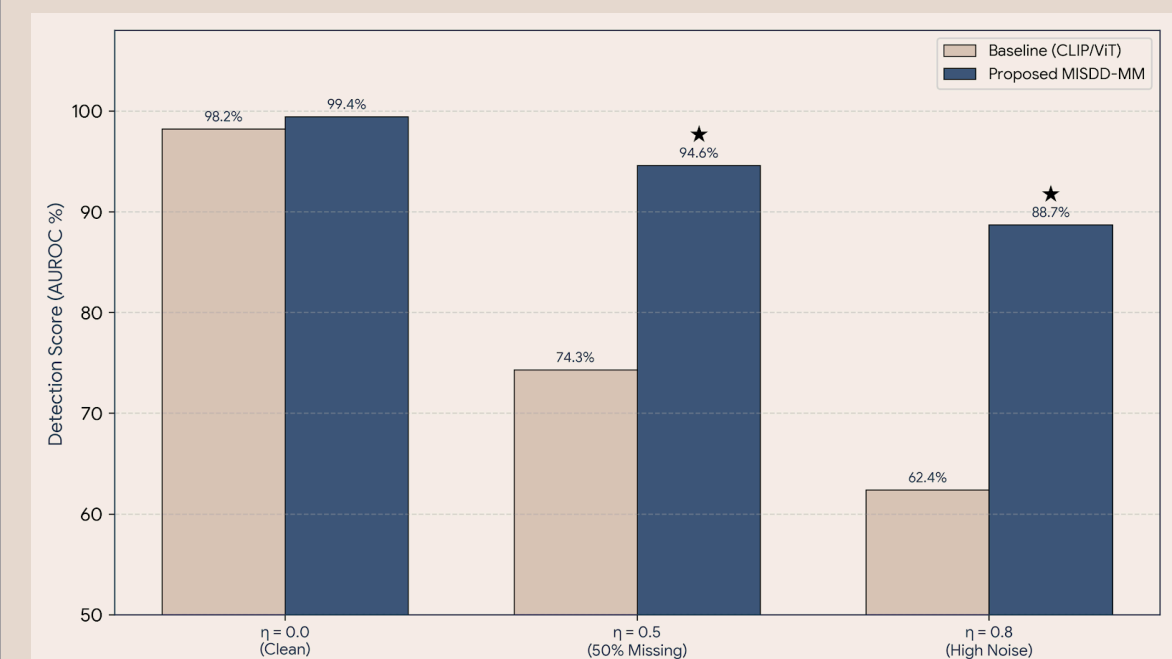


Fig-4: Analysis.

FUTURE WORK

- Deploy **proactive diagnostics** for reliability-aware sensor fusion.
- Integrate **geometric features** to maintain robustness during visual loss.
- Develop alignment protocols for **asynchronous industrial data streams**.
- Optimize model complexity for **real-time manufacturing deployment**.

REFERENCES

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